Sentiment Classification and Temporal Analysis under Processing Constraints

An empirical study using TweetEval and Bluesky

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Abstract — This study evaluates sentiment classification and temporal aggregation under local processing constraints. VADER, a TF-IDF logistic-regression baseline, a bidirectional LSTM and a Twitter RoBERTa checkpoint are compared on 12,284 labeled TweetEval test examples. RoBERTa achieves 72.18% accuracy and 0.724 macro F1, compared with 59.47% and 0.558 for the saved BiLSTM predictions, 59.22% and 0.560 for the linear baseline, and 53.00% and 0.529 for VADER. A separate corpus of 127,741 Bluesky posts returned by an AI search supports temporal analysis over fourteen days. VADER and RoBERTa produce opposite net-sentiment signs in 327 of 336 hourly periods. Repeated CPU inference measurements and a controlled replay through a bounded queue and SQLite storage characterize processing costs and overload behaviour. The evidence shows that classifier selection changes both benchmark performance and the interpretation of aggregate sentiment. Bluesky correctness remains unmeasured because no independent labels are available. The study contributes a reproducible empirical evaluation with explicit sampling, execution and inference boundaries.

Keywords

Sentiment analysis, model comparison, Bluesky, TweetEval, processing latency.

I Introduction

Sentiment monitors transform post labels into temporal signals under processing constraints. This study asks how practical model choice changes benchmark scores, aggregate sentiment and local processing behaviour. It compares VADER, a linear baseline, BiLSTM and Twitter RoBERTa on a common labeled test split, then examines three models on a historical Bluesky search corpus.

II Related Work

VADER provides a rule-based social-media baseline [1]. LSTM and bidirectional recurrent networks motivate the BiLSTM implementation [2, 3]. BERT and RoBERTa establish contextual pretraining approaches whose training conditions influence comparisons [4, 5]. TweetEval supplies the labeled sentiment task [6], while TimeLMs and the Cardiff NLP checkpoint provide Twitter-adapted models [7, 8]. Bluesky infrastructure and analysis systems already support public-data research [9–11]. Our contribution is a bounded empirical evaluation linking model outputs and processing conditions.

III Experimental Design

The local Twitter splits contain 45,615 training, 2,000 validation and 12,284 test rows. All evaluated model predictions refer to the same test rows. The linear model uses 50,000 TF-IDF unigram/bigram features and logistic regression. The BiLSTM configuration uses 100-token inputs, 64-dimensional embeddings and 64 units in each direction. Its saved predictions are evaluated because application control blocks fresh TensorFlow inference.

RoBERTa uses the cached Cardiff NLP checkpoint at revision 3216a57f2a0d9c45a2e6c20157c20c49fb4bf9c7, raw local text and 128-token truncation. VADER uses compound thresholds ±0.05. Accuracy, macro F1 and macro recall describe classification. Paired row-bootstrap intervals use 1,000 resamples and seed 42, assuming independent examples.

The historical AI search retained 127,741 cleaned posts over 19 October–1 November 2026 UTC. It used 15-minute requests capped at 100 results. All 1,344 requests succeeded, while 194 reached the cap. No independent Bluesky labels are available. Predictions align by URI and text. Net sentiment is the positive-minus-negative count divided by total count, aggregated at 15-minute, hourly and daily intervals.

Warm inference timing uses 96 fixed test posts and three repeats at batch sizes one, eight and 32 on a CPU with four PyTorch threads. A separate replay uses 96 cleaned Bluesky posts, one worker, a queue capacity of 16, offered rates of 10, 40 and 80 posts/s, and three repeats. The worker calls the project’s SQLite insert function. Replay omits network delivery and filtering.

IV Results

On the common labeled test split, RoBERTa reaches 72.18% accuracy and 0.724 macro F1. The saved BiLSTM achieves 59.47% and 0.558, the linear baseline 59.22% and 0.560, and VADER 53.00% and 0.529. The neutral baseline reaches 48.33% accuracy. RoBERTa’s 95% bootstrap accuracy interval is 71.38%–72.95%. The paired linear-minus-BiLSTM accuracy interval spans −0.91 to +0.42 percentage points.

On the historical corpus, three-way agreement is 33.53%. RoBERTa and BiLSTM have the highest pairwise agreement at 65.31%, with kappa 0.379. VADER assigns 42.42% positive labels, RoBERTa 11.44% and BiLSTM 13.80%. These distributions describe model outputs rather than validated public sentiment.

VADER and RoBERTa have opposite net-sentiment signs in 327 of 336 hours. Their mean absolute hourly difference is 0.346 and correlation is 0.460. Excluding cap-hit hours

leaves 183 periods and a mean absolute difference of 0.351. Thus the discrepancy persists under this exclusion rule. Six-hour trailing means reduce short-term variation but preserve the model-dependent offset.

Warm batch-one RoBERTa has a median run p50 of 26.45 ms and p95 of 34.61 ms on the fixed test sample. Batch-eight throughput has a median of 59.60 posts/s. These measurements exclude queueing and storage. Replay evaluates those costs on a different, longer-text workload.

VADER replay outcomes were 288/288 stored at 10 posts/s, 288/288 stored at 40 posts/s, 288/288 stored at 80 posts/s. Successful-post latency excludes dropped arrivals, which are retained separately in the experiment records.

RoBERTa replay outcomes were 191/288 stored at 10 posts/s, 79/288 stored at 40 posts/s, 72/288 stored at 80 posts/s. Successful-post latency excludes dropped arrivals, which are retained separately in the experiment records.

V Discussion and Limitations

A common sample can produce different aggregate sentiment directions depending on its classifier. Benchmark correctness, inter-model consistency and local runtime answer distinct empirical questions. The results support reporting all three when describing a monitor. Pretraining histories differ, so the comparison evaluates model instances rather than isolating architectural effects.

One query, a two-week window and capped responses limit representativeness. Whole-post polarity does not establish stance toward AI. Bluesky correctness is unmeasured. Bootstrap intervals omit training-seed and cluster dependence. BiLSTM timing is unavailable. Short replay runs exclude the network and production orchestration, and cannot establish sustained capacity.

VI Conclusion

RoBERTa leads the local labeled benchmark, while the linear baseline and saved BiLSTM have similar scores within paired uncertainty. Historical temporal estimates depend strongly on classifier choice. Controlled replay adds evidence about processing constraints. The project supplies code, prediction files and timing observations for a reproducible empirical study.

References

[1] C. J. Hutto and E. Gilbert, “VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text,” Proc. ICWSM, vol. 8, no. 1, pp. 216–225, 2014. doi: 10.1609/icwsm.v8i1.14550.

[2] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.

[3] M. Schuster and K. K. Paliwal, “Bidirectional Recurrent Neural Networks,” IEEE Transactions on Signal Processing, vol. 45, no. 11, pp. 2673–2681, 1997. doi: 10.1109/78.650093.

[4] J. Devlin, M.-W. Chang, K. Lee and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” Proc. NAACL-HLT, pp. 4171–4186, 2019. https://aclanthology.org/N19-1423/

[5] Y. Liu et al., “RoBERTa: A Robustly Optimized BERT Pretraining Approach,” arXiv:1907.11692, 2019. https://arxiv.org/abs/1907.11692

[6] F. Barbieri, J. Camacho-Collados, L. Espinosa Anke and L. Neves, “TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification,” Findings of EMNLP, pp. 1644–1650, 2020. https://aclanthology.org/2020.findings-emnlp.148/

[7] D. Loureiro, F. Barbieri, L. Neves, L. Espinosa Anke and J. Camacho-Collados, “TimeLMs: Diachronic Language Models from Twitter,” Proc. ACL System Demonstrations, pp. 251–260, 2022. https://aclanthology.org/2022.acl-demo.25/

[8] Cardiff NLP, “twitter-roberta-base-sentiment-latest,” model card, accessed Nov. 9, 2026. Evaluated checkpoint revision 3216a57f2a0d9c45a2e6c20157c20c49fb4bf9c7. https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest

[9] M. Kleppmann et al., “Bluesky and the AT Protocol: Usable Decentralized Social Media,” DIN 2024. https://arxiv.org/abs/2402.03239

[10] G. Chhetri, A. Dutta and S. Das, “CognitiveSky: Scalable Sentiment and Narrative Analysis for Decentralized Social Media,” arXiv:2509.11444, 2025. https://arxiv.org/html/2509.11444v1

[11] U. Jeong, B. Jiang, Z. Tan, H. R. Bernard and H. Liu, “BlueTempNet: A Temporal Multi-network Dataset of Social Interactions in Bluesky Social,” arXiv:2407.17451, 2024. https://arxiv.org/abs/2407.17451